Arduino Exercise

This exercise is to be undertaken individually. The task is detailed below. You should do the design using an Arduino with the sensors, servos and other components you have access to. It is fine to use code from the Arduino website or other sources, but make sure that any code you make use of you properly attribute its use. You may wish to prototype your work using Tinkercad as a simulator.

Task

In this task you will implement an Arduino based solution for a distance sensor that could be used by a robot when approaching an object. The system is to use an ultrasonic distance sensor to measure the distance to the object. Then using on the distance measured as a variable move a servo between 0 and 90 degrees with 0 degrees corresponding to a range of 15cm and 90 degrees corresponding to 195cm. In addition you should have at least one other means of indicating the range, for example using a LCD display, or a pulsing LED or a varying tone from a piezo sounder. It is up to you which other ways of indicating the range you use, in your video make sure you explain how the methods you have chosen indicate the distance of the object.

There are two major phases. The first is the design phase, which consists of reading through the exercise, planning, and coming up with a design. The next phase is to implement the task using the Arduino software and a simulator or an actual Arduino board.

You are to produce a short video of the task, the video should be between 1 and 3 minutes in duration.

Guidelines for Video

The video should show your circuit with an actual Arduino. It should also show you testing the circuit to demonstrate its operation. You should also upload the text file of the code that you have used for the Arduino. Upload a single compressed file that includes your video and code as detailed below.

Use your email ID as a file name, for example, "ma722.zip" If you face issues uploading your file due to large size, use MS Stream to host your video, make sure that you add the project manager in your campus as owner in the video settings. Then include the video link in a document that includes your code to upload here.